Linear relationships

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- creates and displays number patterns and finds graphical solutions to problems involving linear relationships **MA4-LIN-C-01**

Related Life Skills outcomes: MALS-POS-01

Content

Plot and identify points on the Cartesian plane

- Plot and label points on the Cartesian plane of given coordinates, including those with coordinates that are not whole numbers

Example(s):

Plotting and labelling the points $P\left(\frac{1}{3}, -2\right)$ and $Q(-3.5,0)$ on the Cartesian plane.

Aboriginal art uses topographical views to map Country. Use a topographical map of the local region to identify coordinates of landmarks such as watering holes, cultural places or places of personal significance.

NPA8

- Identify and record the coordinates of given points on the Cartesian plane, including those with coordinates that are not whole numbers

Example(s):

Using the diagram of the Cartesian plane to identify the coordinates of point *A*, naming the point at $\left(4,\ -2\frac{1}{2}\right)$, labelling point *Z* on the x-axis directly above point *C*, and identifying the coordinates of the point $2\frac{1}{2}$ units left of the origin.

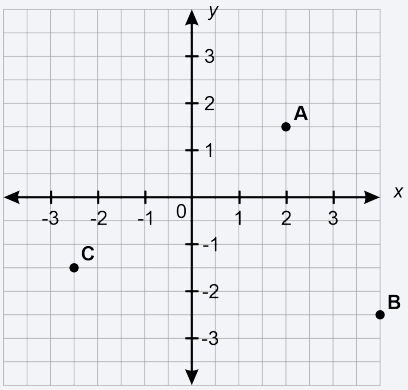


Image long description: A number plane with an x and y-axis. Point A is plotted at 2, 1.5. Point B is plotted at 4, minus 2.5. Point C at minus 2.5, minus 1.5

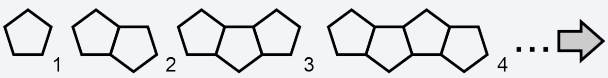
PoL5

Plot linear relationships on the Cartesian plane

- Construct a geometric pattern and record the results in a table of values

Example(s):

Representing the pattern below made from matches in a table of values:



number of pentagons *(p)*	1	2	3	4	...
number of matches *(m)*	5	9		17	...

Image long description: The number of pentagons is measured against the number of matches used to make the pentagon pattern, represented in a table. The table shows the number of pentagons on the top row (1, 2, 3, 4) and the corresponding number of matches on the bottom row (5, 9, blank, 17).

- Represent a given number pattern (including decreasing patterns) using a table of values
NPA5
- Describe a number pattern in words and generate an equation using algebraic symbols

Example(s):

Describing a number pattern as: the y-value is double the x-value, and generalising the equation to be $y = 2x$.

NPA7

- Apply an equation generated from a pattern to calculate the corresponding value for a smaller or larger number

Example(s):

Generating the equation for the pattern and completing the table of values.

x	1	2	3	4	5	8	95
y	-13	-15	-17				

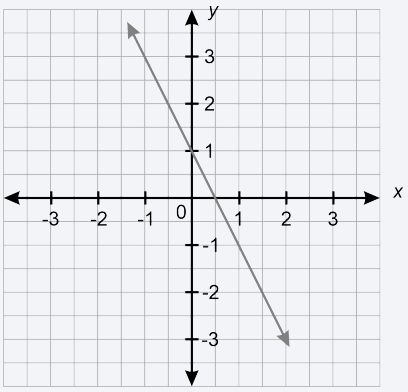
Image long description: A table with x-values on the top row (1, 2, 3, 4, 5, 8, 95) and y-values on the bottom row (minus 13, minus 15 and minus 17). The remaining values for y are left blank.

NPA7

- Recognise that a linear relationship can be represented by a number pattern, an equation (or a rule using algebraic symbols), a table of values, a set of pairs of coordinates and a line graphed on a Cartesian plane, and move flexibly between these representations

Example(s):

Deriving the equation representing the relationship between x and y by completing a table of values for the set of points shown in the diagram.



NPA7, NPA8

- Explain that there are an infinite number of ordered pairs that satisfy a given linear relationship by extending a line joining a set of points on the Cartesian plane
- Compare similarities and differences of multiple straight-line graphs on the same set of axes using graphing applications

Example(s):

Using digital tools to graph $y = x + 1$, $y = x - 2$ and $y = -2x + 1$ on the same Cartesian plane, identifying similarities and differences such as *parallel*, *non-parallel*, *pass through the same point*, *increasing* and *decreasing.*

- Describe linear relationships in real-life contexts and solve related problems

Example(s):

Demonstrating a linear relationship using a graph for a situation where an electrician charges a $75 call-out fee and $60 for each hour it takes to complete a job.

NPA8, IRD6

Solve linear equations using graphical techniques

- Recognise that each point on the graph of a linear relationship satisfies the equation of a line

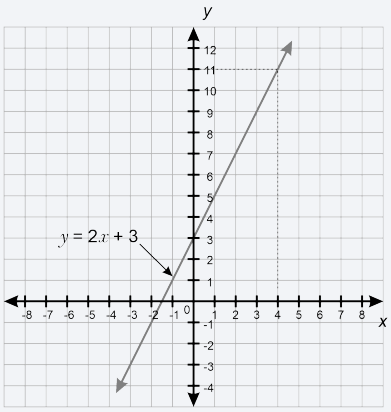
Example(s):

Verifying that the points $(2,\ 5)$ and $(-1,\ -1)$ satisfy the line $y = 2x + 1$.

- Apply graphs of linear relationships to solve a corresponding linear equation using graphing applications

Example(s):

Using the graph of $y = 2x + 3$ to find the solution to the equation $2x + 3 = 11$.

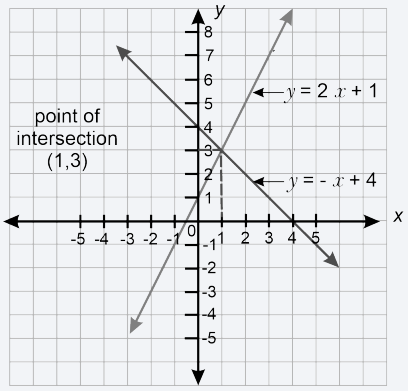


IRD6

- Graph 2 intersecting lines on the same set of axes and identify the point of intersection using either graphing applications or a table of values

Example(s):

Graphing the linear equations $y = 2x + 1$ and $y = -x + 4$ and identifying their point of intersection.



- Verify that the point of intersection satisfies the equations of both lines